Exercise – *Croquet*

"Alice thought she had never seen such a curious croquet-ground in her life; it was all ridges and furrows."

Alice's Adventures in Wonderland *by Lewis Carroll (1832–1898)*

At the court of the Queen of Hearts they play a peculiar game of croquet. The goal is to move a ball to a given target location t, using as few shots as possible. Each *shot* consists of hitting the ball with a club, thereby moving the ball through the air by a distance of at most d from its current location. As the playing grounds are very uneven, only a few select locations are *safe* for the ball. If the ball is shot to a location that is not safe, then the ball and hence the game is lost—a situation that should be avoided. In contrast, the starting location is not required to be safe, as Alice may setup the ball on a tee[1] for the first shot. For a given starting location s and a given number k of shots, can Alice move the ball from s to t using at most k shots?

Input The first line of the input contains the number $t \leqslant 30$ of test cases. Each of the t test cases is described as follows.

- It starts with a line containing four integers n m q k, separated by a space. They denote
  - n, the number of safe locations on the playing grounds ($2 \leqslant n \leqslant 4 \cdot 10^4$);
  - m, the number of different starting locations ($1 \leqslant m \leqslant 4 \cdot 10^4$);
  - $q = d^2$, where d is the distance that a single shot can bridge at most ($1 \leqslant q \leqslant 2^{53}$);
  - k, the maximum number of shots allowed.
- The following n lines define the safe locations. The last safe location is the target location t.
- The following m lines define the different starting locations s_i, for $i \in \{0, \ldots, m-1\}$.

Each location is described as a point in $\mathbb{R}^2$, by two integer coordinates x y, separated by a space and such that $|x|, |y| < 2^{24}$. You may assume that the n safe locations are pairwise distinct.

Output For each test case output a line with one character "y" or "n" per starting point, that is, a string $c_0 c_1 \cdots c_{m-1}$ of m characters. For $i \in \{0, \ldots, m-1\}$, the character c_i is "y" if and only if Alice can move the ball from s_i to t using at most k shots.

Points There are four groups of test sets worth 80 points in total. For each group there is also a corresponding hidden test set that is worth 5 points. So, you can achieve $80 + 4 \cdot 5 = 100$ points overall.

1. For the first group of test sets, worth 25 points, you may assume that $k = 2$.
2. For the second group of test sets, worth 25 points, you may assume that $k = n$.
3. For the third of test sets, worth 20 points, you may assume that $k \leqslant 20$.
4. For the fourth group of test sets, worth 10 points, there are no additional assumptions.

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[1] A stand used in sport to support and elevate a stationary ball prior to striking with a foot, club or bat.

Corresponding sample test sets are contained in `testi.in/out`, for $i \in \{1, 2, 3, 4\}$.

Sample Input

```
1
3 4 2 2
1 1
2 1
0 0
3 1
2 0
-1 0
-2 0
```

Sample Output

```
nyyn
```